

TwistBench: Benchmarking Transformational Creativity in LLMs via Literary Plot Twists

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Abstract

Benchmarks and tests of creativity for Large Language Models (LLMs) are now abundant—it has been shown that LLMs can realize aspects of combinatorial creativity, exploratory creativity, and that they exceed human performance on psychometric tests of creativity. However, these evaluations have failed to address more complex types of creativity recognized in cognitive science, such as *transformational creativity*. Widely held to underpin major scientific, artistic, and technological breakthroughs, transformational creativity describes the ability to *radically modify an existing conceptual space*, such as a scientific paradigm, a musical genre, or, as we show, the plot of a short story. To resolve this gap in the benchmarking landscape, we introduce **TwistBench**, a benchmark of literary plot twists for assessing transformational creativity. We collect a set of stories with plot twists written by famous human authors and compare them against stories written by 71 LLMs, across scoring dimensions *diversity*, *surprise*, *coherence*, and *realism*. Our main findings are that: (1) Even the latest frontier LLMs underperform expert humans in transformational creativity, though elaborate prompting strategies like in-context regeneration can raise the performance of select models to match expert humans. (2) Two dominant failure modes plague LLMs when applied to open-ended, transformative tasks: (a) *mode collapse*, where frontier models are capable of generating high quality twists, but these twists are largely the same and lack diversity; and (b) *breaking the world model*, where models give surprising and coherent plot twists, but do so in a way that measurably departs from physical reality (e.g. time travel, ghosts), violating literary “fair play.” (3) An analysis of the creative processes employed by reasoning models reveals process-level homogeneity among LLMs. Namely, models always choose the plot twist *first*, then retroactively determine the plot conditioned on the twist. While today’s LLMs struggle with transformational creativity at the level of expert humans, future work can use **TwistBench** as a testbed to design new methods and techniques that enhance this important ability.

1 Introduction

Creativity has long been heralded as the pinnacle of human intelligence (Simonton, 2001). It is no surprise, then, that considerable effort has gone towards assessing the creativity of large language models (LLMs) in recent years (Ismayilzada et al., 2024). A growing body of work shows that LLMs can realize forms of creativity, such as combinatorial creativity (Gu et al., 2025) or divergent thinking (Guilford, 1956), at or exceeding human levels (Bellemare-

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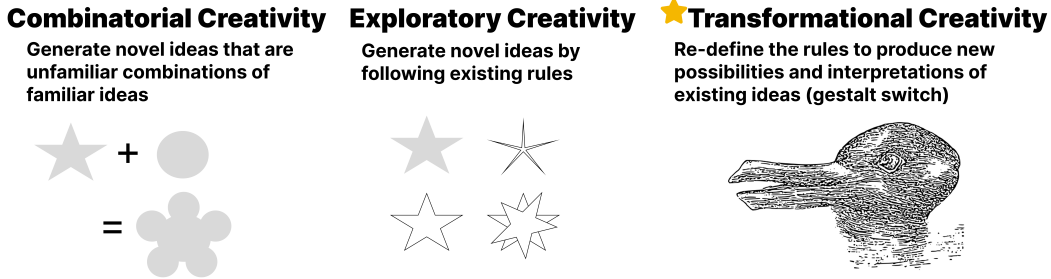


Figure 1: **Boden’s three types of creativity** (Boden, 2004): *combinatorial* (novel combinations of familiar ideas), *exploratory* (novel artifacts within a fixed conceptual space), and *transformational* (restructuring the space itself). We benchmark the last via literary plot twists.

Pepin et al., 2024; Stevenson et al., 2022; Si et al., 2024), but may fail to surpass humans at more complex types of creativity, especially in scientific domains (Si et al., 2025).

Indeed, different forms of creativity vary widely in the complexity required to realize them. In the cognitive science literature, a popular categorization of creative processes is Boden’s (2004) distinction between combinatorial, exploratory, and transformational creativity. *Combinatorial creativity* involves making unfamiliar combinations of familiar ideas, such as combining ingredients to form a new recipe (Varshney et al., 2019).¹ *Exploratory creativity*, meanwhile, presupposes an existing set of rules and constraints, called a conceptual space, that one can follow to discover or create something new, such as inventing a new chess opening (Boden, 2004).

Lastly, *transformational creativity* is the most radical but complex type of creativity, one that involves deliberate restructuring of a conceptual space (Wiggins, 2006; Schapiro et al., 2025). A clear example of this is the invention of non-Euclidean geometry, in which a single assumption (the parallel postulate) was modified to elicit a radically transformed space (Schapiro et al., 2025). Although transformational creativity has historically underpinned major scientific, technological, and artistic breakthroughs (Kuhn, 1970; Thagard, 2018), it has been overlooked by the LLM benchmarking landscape due to the challenge of reducing such a complex ability into a self-contained and measurable task (Nagarajan et al., 2025). At small scale, LLMs have begun to be assessed on transformative *generalization* during mathematical problem solving (Sun et al., 2025), though this setting departs from core desiderata for creativity evaluation like open-endedness (Soros et al., 2024).

The present study Resolving this gap in the benchmarking landscape, we establish the first large-scale benchmark of Transformational Creativity for LLMs, challenging models to generate a *diverse* set of *surprising*, *coherent*, and *realistic* literary plot twists (**TwistBench**). Literary plot twists require modifying or restructuring conceptual elements established by the story, satisfying the definition of transformational creativity as modifying an existing conceptual space (Boden, 2004; Wiggins, 2006; Schapiro et al., 2025). In particular, writing a story itself supplies the conceptual space, and then the plot twist is a

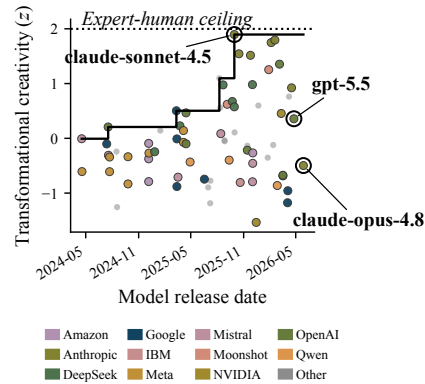


Figure 2: **Frontier models underperform expert humans in transformational creativity.** Each model’s score is the mean of its z-scored *surprise*, *gated coherence*, and *diversity*. Scores have risen over time but none reaches the expert-human ceiling.

¹Combinatorial creativity dates back to as early as Jacques Hadamard’s (1954) survey of mathematicians and physicists, in which Einstein famously said that “combinatory play seems to be the essential feature in productive thought.”

single, unambiguous event that restructures the conceptual space. Afterwards, stories can then be evaluated according to standard evaluation criteria for creative products that assess novelty, surprise, and utility (Maher, 2010; Simonton, 2010). We administer **TwistBench** to a large set of 71 LLMs and find that humans rank first overall above all 71 LLMs tested, though for one model, elaborate prompting strategies such as in-context regeneration raise its performance to match human scores.

Main contributions In detail, our main contributions and findings are as follows:

1. In Section 3, we introduce **TwistBench**, a benchmark of literary plot twists for assessing transformational creativity. To the best of our knowledge, **TwistBench** is the first large-scale transformational creativity benchmark for LLMs.
2. In Section 4, we find that expert human authors outperform LLMs in writing a diverse set of surprising, coherent, and realistic stories with plot twists, ranking first overall above all 71 LLMs tested. The failure modes of LLMs can be neatly categorized according to two types: #1 (*mode collapse*), where models capable of producing high quality stories, but not multiple diverse ones; and #2 (*breaking the world model*), where models give surprising but unrealistic twists involving supernatural elements that depart from physical reality.
3. In Section 4.2, using elaborate prompting strategies such as in-context regeneration, we raise the performance of a single model (Claude Opus 4.5) to match human scores, indicating careful use of test-time compute can improve more complex types of creativity among select models.
4. In Section 5, we analyze the creative processes employed by reasoning models by analyzing thinking traces. We find that even as thinking is increased from low to high, this process remains remarkably homogeneous, with models always choosing the twist *first*, before retroactively determining the plot.

Overall, our findings reveal that today’s LLMs struggle with transformational creativity at the level of expert humans, suffering from failure modes like mode collapse and a tendency to give unrealistic twists. However, all hope is not lost: using in-context regeneration, we raised the performance of Claude Sonnet 4.5 to match expert humans, suggesting that principled use of test-time compute can enhance more complex types of creativity among LLMs, and in select cases, raise performance to the level of expert humans. Future work can use **TwistBench** as a testbed to design new methods and techniques that enhance this important ability.

2 Background and Related Work

Before introducing our benchmark, we start by providing background and related work. Although many types of creativity are recognized in cognitive science, some remain un-addressed among LLM benchmarks. Here, we review classifications of creativity types, underscoring the gaps that motivate the present study.

Types of Creativity Creativity can be attributed to a **p**erson, **p**rocess, **p**roduct, or **p**ress² (cf. the 4 P’s of creativity: Rhodes (1961)). Many frameworks for categorizing creative **p**rocesses and **p**roducts have been proposed in cognitive science, such as Boden’s (2004) distinction between combinatorial, exploratory, and transformational creativity; the Four-C model of creativity (Kaufman & Beghetto, 2009) describing personal creativity (Mini-c), everyday creativity (Little-c), professional creativity (Pro-c), and eminent creativity (Big-C); or the distinction between divergent and convergent thinking (Guilford, 1956; Dietrich, 2004). Here, we focus on Boden’s (2004) threefold taxonomy (Figure 1), now increasingly sourced as inspiration for LLM creativity benchmarks and tasks (Nagarajan et al., 2025; Wadhwa et al., 2026; Sun et al., 2025; Schapiro et al., 2026b). (i) *Combinatorial creativity* refers to the process of making unfamiliar combinations of familiar ideas, which is commonly invoked during wordplay (Nagarajan et al., 2025), scientific discovery (Simonton, 2021), and technological

²Press here refers to an environment or zeitgeist (Simonton, 2004).

innovation (Thagard, 2010). Structured forms of combinatorial creativity include conceptual blending (Fauconnier & Turner, 2008) and bisociation (Koestler, 1964). Meanwhile, (ii) *exploratory creativity* refers to the process of generating novel yet admissible artifacts without disturbing the rules that define a conceptual space, such as when a composer writes within an established genre or a scientist works within a prevailing paradigm (Boden, 2004; Wiggins, 2006; Kuhn, 1970). Finally, (iii) *transformational creativity* describes the most complex type of creativity in which a conceptual space is restructured, and old artifacts come to be viewed in an entirely new light (Wiggins, 2006; Schapiro et al., 2025). Unlike exploratory creativity that achieves novelty by following existing rules, transformational creativity redefines the *rules themselves* to provide a novel interpretation of existing ideas, invoking what many consider to be a perceptual-like gestalt switch (Boden, 2004; Kuhn, 1970) (cf. Figure 1).

Benchmarks and Tests of LLM Creativity Benchmarks of LLM creativity commonly assess constructs such as divergent thinking (Zhang et al., 2025; Jiang et al., 2025; Qian et al., 2025), scientific ideation ability (Ruan et al., 2026), and creative writing (Mazur, 2025; Chiang et al., 2024; Paech, 2023). Among existing creative writing benchmarks, it is common to apply LLM-as-a-judge evaluation with fixed rubrics assessing qualities such as fluency, narrative coherence, characterization, imagery, and emotional impact (Mazur, 2025), or to elicit pairwise human preferences over model-written passages (Chiang et al., 2024). Meanwhile, other studies have administered human psychometric tests—such as the Divergent Association Task (Olson et al., 2021), the Remote Associates Test (Mednick, 1962), the Alternative Uses Test (Guilford, 1956), and the Torrance Tests of Creative Thinking (Torrance, 1974)—to LLMs, enabling direct human-AI comparison. On psychometric tests in particular, LLMs have been found to match or exceed human creativity (Stevenson et al., 2022; Bellemare-Pepin et al., 2024). However, in a two-part study comparing humans and LLMs on scientific ideation, Si and colleagues (2024; 2025) found that while LLMs can match or exceed the novelty of human-generated ideas, LLM-generated ideas are often less practically feasible or experimentally lucrative, in what has been termed the ideation-execution gap. Together with a documented tendency of LLMs to homogenize in open-ended domains (Jiang et al., 2025; Zhang et al., 2025) known as the artificial hivemind effect, these failures in more complex creative tasks challenge whether current models can do more than merely explore or combine within an existing space—and instead deliberately *transform* one.

3 Benchmarking Transformational Creativity via Literary Plot Twists

Although progress in designing an LLM benchmark of transformational creativity has been stifled by the challenge of reducing such a complex ability into a self-contained and measurable task (Nagarajan et al., 2025), here we show that literary plot twists satisfy the definitions laid forth by Boden (2004) and advanced in subsequent studies (Wiggins, 2006; Schapiro et al., 2025). Namely, writing the story itself supplies a conceptual space, and then the plot twist is a single discrete artifact that restructures the conceptual space. Afterwards, the story can be evaluated according to whether the restructuring was realistic and maintains logical coherence, mapping cleanly onto standard evaluation criteria for creative products like novelty, surprise, and utility (Maher, 2010; Simonton, 2010). We start with a motivating example (§ 3.1) before introducing the benchmark (§ 3.2).

3.1 A Motivating Example

Consider a story as follows: (e1) a man sits at a hospital bedside night after night, (e2) holding the hand of the woman he loves, (e3) reciting the memories of their life together, and (e4) willing her to wake. (e5) She stirs, (e6) her eyes find his, and (e7) the two weep as recognition returns. Here, the plot establishes a conceptual space (cf. Boden (2004)) of a man visiting his bedridden wife in the hospital, consisting of implicit assumptions that (a1) the man is her partner and (a2) the memories are theirs.

Now, the story continues: (e8) a nurse gently tells him that visiting hours are over and (e9) walks him back to his own room down the hall. Then, it is revealed that **(e10)**

Table 1: **An example of transformational creativity** (Section 3.1). Each assumption, inferred from earlier events, is overturned by a *realistic* plot twist that forces those events to be re-interpreted (*surprise*) while remaining logically consistent (*coherence*).

Implicit assumption	Inferred from	Overturned by the twist
(a1) the man is her partner	e2, e7	(e10) he is a patient; the woman is a stranger
(a2) the memories are theirs	e3	(e11) the memories are lines from the day-room television

he is a patient on the same ward, the woman is a stranger, and (e11) the memories he recites each night are lines from the television that plays each day. Woah! The events e9, e10, e11 contain a plot twist that transforms the interpretation of earlier events (e2, e3, e7) and overrides inferences drawn from them, as summarized in Table 1.

Note, however, that the plot twist only works because it meets two further conditions. (1) It is *coherent*: nothing earlier need be retracted, as the same details that first read as devotion now read as confusion. (2) It is *realistic*: the surprise comes from *recontextualizing* the world the reader was already given, not from importing new rules into it—the notion of literary “fair play” (Hühn, 1987; Wagner et al., 2025). This second condition is what keeps the move *transformational* rather than merely disruptive: transformational creativity restructures a conceptual space while keeping the result coherent and intelligible (Boden, 2004; Wiggins, 2006), and a genuine twist does so by recasting the very elements the story established. Resolving the story by declaring the ward a dream or a simulation is also surprising, but it does not restructure that space—it discards it for a new one with different rules. Such moves can be imaginative in their own right, yet they are a *distinct* operation, and they admit a trivial shortcut: *any* story can be made “surprising” by revealing that none of it was real. We therefore require realism so that surprise is earned by transforming the world the story has committed to, not by escaping it.

3.2 Benchmark Setup

Our benchmark assesses the ability for LLMs to write a *diverse* set of stories with *surprising*, *coherent*, and *realistic* plot twists.

Models and Inference We evaluate a large set of 71 LLMs from 19 providers, ranging from small open-weight models (e.g. Llama-3.2-3B) to the latest frontier systems (e.g. Claude Opus 4.8, GPT-5.5). Each model is asked to write a complete short story containing a plot twist, as in Figure 3. We sample each model at three temperatures (0.9, 1.0, 1.2) with ten samples each (up to 30 stories per model, averaged across temperatures during overall scoring) and constrain the length to 2,000–3,000 words³

Metrics Creative products are typically evaluated according to their surprise, novelty, and utility (Varshney, 2019; Maher, 2010; Simonton, 2010). In the context of literary surprise, specifically, it is standard to invoke scoring dimensions surprise, coherence, consistency, and divergence (Chieppe et al., 2022), which we broadly adopt as inspiration here. Given a story T , we assess the following dimensions via Likert scales (Likert, 1932):

1. **Surprise:** $\text{Surp}(T) \in \{1, 2, 3, 4, 5\}$. How drastically must one reinterpret earlier story elements after the plot twist?
2. **Coherence:** $\text{Coh}(T) \in \{1, 2, 3, 4, 5\}$. After the plot twist, do earlier story elements still hold together? Is the story still logically consistent?
3. **Realism:** $\text{Real}(T) \in \{1, 2, 3, 4, 5\}$. Does the plot twist involve impossible elements (e.g. real ghosts, magic, time travel, sentient AI, aliens) that violate the notion of literary fair play?

³This range surrounds the $\sim 2,500$ -word median of the human stories we define and introduce at the end of Section 3, mitigating story length as a confounding variable.

Task and Prompt: Write a Short Story With a Plot Twist

Write a complete short story (roughly 2,000-3,000 words). The story must contain a plot twist: a late reveal that makes the reader re-interpret earlier events in a new light, yet is consistent with everything that came before. In hindsight it should feel prepared, not arbitrary. Choose your own subject and characters. Do not announce or explain the twist. Write only the story, no title or commentary.

Figure 3: **Task and Prompt.** The instructions used to administer **TwistBench**.

To score each dimension in practice, we use a three-judge ensemble (Claude Sonnet 4, GPT-4o, and Gemini 2.5 Flash) aggregated per dimension by the median and using a fixed rubric, given in Section B. Moreover, to mitigate self-preference bias in LLM-as-a-judge (Wataoka et al., 2024), the three judges are separate from the 71 LLMs evaluated as part of the benchmark results. The reliability of the three judges over all $n = 2,035$ stories, as given by the intra-class correlation coefficient ICC(2,k) (Shrout & Fleiss, 1979), is good-to-excellent on every dimension: realism at 0.91, surprise at 0.88, and coherence at 0.79.

Next, since diversity is a crucial aspect of creativity (Nagarajan et al., 2025) and given that LLMs have a documented tendency to homogenize on open-ended tasks (Zhang et al., 2025; Jiang et al., 2025), we also measure diversity over the set of a model’s plot twists. Given a set of n stories $\mathcal{T} = \{T_1, T_2, \dots, T_n\}$ and a function that maps a story to a concise summary of its twist $f : T \rightarrow W^4$, we compute the average embedding dissimilarity among these twists:

$$\text{Div}(\mathcal{T}) := \frac{1}{n(n-1)} \sum_{i \neq j}^n d_{\cos}(\phi(f(T_i)), \phi(f(T_j))) \quad (1)$$

where ϕ refers to a text embedding model.⁵

Overall Score Consistent with prior tests and benchmarks of LLM creativity that reward novelty only when it is sufficiently appropriate (Nakajima et al., 2026; Wadhwa et al., 2026; Schapiro et al., 2026a), we treat realism as a constraint, rather than its own separate dimension, such that a story’s surprise and coherence only count when the story is fully realistic (i.e., $\text{Real}(T) = 5$):

$$S^*(T_i) := S(T_i) \mathbb{I}[\text{Real}(T_i) = 5], \quad \text{Coh}^*(T_i) := \text{Coh}(T_i) \mathbb{I}[\text{Real}(T_i) = 5]. \quad (2)$$

Then, given a set of n stories $\mathcal{T}$ generated by a model (or human) m , with mean surprise $\bar{S}_m^* = \frac{1}{n} \sum_i S^*(T_i)$, mean coherence $\bar{C}_m^* = \frac{1}{n} \sum_i \text{Coh}^*(T_i)$, and diversity $\text{Div}_m = \text{Div}(\mathcal{T}_m)$, we define the *transformational creativity* score as the average z-score of each dimension over the pool of all 71 models:

$$\text{TC}(\mathcal{T}) := \frac{1}{3} [z(\bar{S}_m^*) + z(\bar{C}_m^*) + z(\text{Div}_m)]. \quad (3)$$

Expert Human Stories Lastly, to compare against instances of Pro-C (Kaufman & Beghetto, 2009) transformational creativity from expert humans, we collect a set of 18 stories with plot twists written by famous human authors, such as Kate Chopin, Guy de Maupassant, O. Henry, and Mark Twain. The full list is given in Section F, including the length of each story, the location of the plot twist within the text, and the year the story was published.

4 Results

4.1 Key Observations

Expert humans outperform LLMs in transformational creativity Human writers achieve the highest transformational creativity scores at $z = +2.00$ overall, with *surprise*: $+0.78$

⁴Empirically, f is gpt-4o-mini prompted to summarize each story’s twist in one sentence. The full prompt is given in Section C.

⁵In this work, we use all-mpnet-base-v2 (Song et al., 2020).

Table 2: Top systems by transformational creativity. The 8 highest-scoring systems on *Overall* and on each facet. Expert humans (bold) top the *Overall* ranking and reach the top 8 on *Coherence* and *Realistic*, but rank only #9 on *Diversity* and #14 on *Surprise*.

Rank	Overall	Surprise	Coherence	Diversity	Realistic
1	Expert humans (+2.0)	claude-opus-4.8 (+1.4)	claude-opus-4.6 (+1.3)	deepseek-v3.2 (+1.4)	claude-sonnet-4.5 (+1.7)
2	claude-sonnet-4.5 (+1.9)	deepseek-v3.2-exp (+1.3)	claude-opus-4.8 (+1.3)	deepseek-chat-v3.1 (+1.4)	claude-sonnet-4.6 (+1.7)
3	claude-sonnet-4.6 (+1.8)	gemma-4-31b (+1.3)	Expert humans (+1.2)	kimi-k2 (+1.3)	claude-opus-4.5 (+1.7)
4	claude-opus-4.6 (+1.7)	qwen3-next-80b (+1.2)	claude-sonnet-4.6 (+1.2)	deepseek-v3.2-exp (+1.3)	claude-opus-4.6 (+1.7)
5	claude-haiku-4.5 (+1.5)	deepseek-chat-v3.1 (+1.1)	claude-opus-4.7 (+1.2)	deepseek-v3.1-term. (+1.2)	Expert humans (+1.6)
6	claude-opus-4.5 (+1.5)	deepseek-v3.1-term. (+1.0)	deepseek-v3.2-exp (+1.2)	glm-5.1 (+1.2)	mistral-small-3.1-24b (+1.5)
7	gpt-5.4 (+1.4)	qwen3.5-flash (+1.0)	gemma-4-31b (+1.2)	kimi-k2.5 (+1.2)	gpt-5.4 (+1.3)
8	kimi-k2.5 (+1.3)	deepseek-v3.2 (+0.9)	claude-haiku-4.5 (+1.2)	gemma-3-27b (+1.2)	granite-4.0-h-micro (+1.3)
9	jamba-large-1.7 (+1.1)	gpt-5.4-mini (+0.9)	claude-opus-4.5 (+1.2)	Expert humans (+1.2)	claude-haiku-4.5 (+1.3)
14	deepseek-v3.1-term. (+0.7)	Expert humans (+0.8)	glm-5.1 (+1.0)	nova-2-lite-v1 (+0.9)	hunyuan (+1.0)

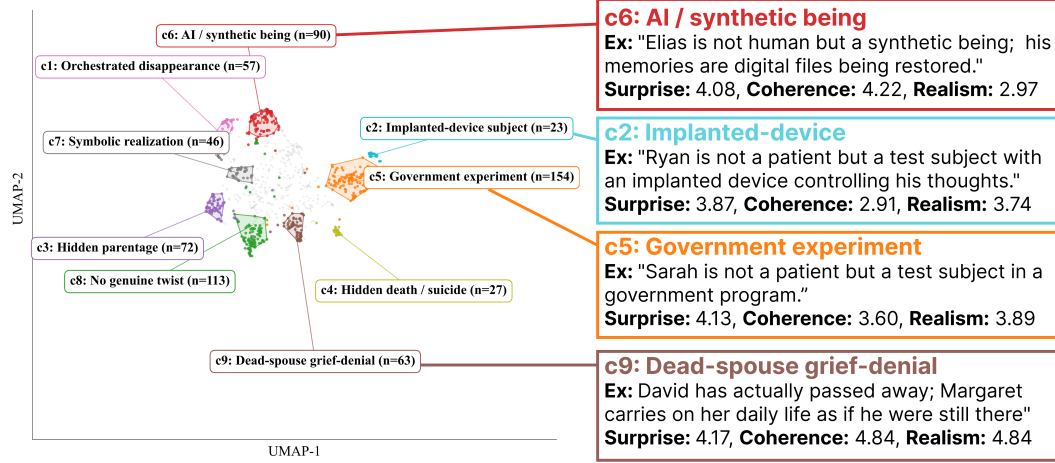


Figure 4: Recurring patterns in LLM-generated plot twists. A map of recurring twists models give, obtained via spectral clustering and projected to 2D with UMAP (McInnes et al., 2020). Surprise, coherence, and realism in each box are the cluster means.

(rank 14/71), *coherence*: +1.25 (rank 3/71), *diversity*: +1.15 (rank 9/71), and *realism*: +1.64 (rank 5/71). Notably, this is not due to winning on any individual dimension, but instead through a balanced profile overall. A small set of frontier models score just below humans in overall scores, with Claude-Sonnet-4.5 at $z = +1.9$ overall, Claude-Sonnet-4.6 at $z = 1.8$, and Claude-Opus-4.6 at $z = +1.7$, but performance quickly drops starting at rank 5/71. We observe two consistent failure modes among models that illustrate why LLMs underperform expert humans.

Failure Mode #1: Mode Collapse Although the Claude Sonnet family scores just under expert human writers, Claude-Opus-4.5 and Opus-4.6 suffer from a lack of diversity, as shown in Figure 5(b). Although these models generate plot twists that are as surprising, realistic, and coherent as human writers, they fail to generate a diverse set of high quality stories, consistent with broader findings that many post-training techniques tend to reduce output diversity (Chen et al., 2026; Li et al., 2025; 2026). Concretely, 10 of Claude Opus 4.5’s 30 stories collapse onto a single reveal cluster, **(c9) dead-spouse grief-denial** (Figure 4): a loved one is secretly long dead and the protagonist carries on in denial. Each story is surprising, coherent, and realistic, but the set of such stories is not diverse. More broadly, Figure 4 shows the full set of recurring patterns among LLM-generated plot twists that can be neatly clustered, including especially unrealistic twists involving **(c6) AI and synthetic beings** that score low on realism. This leads to our next observed failure mode.

Failure Mode #2: Breaking the World Model Transformational creativity involves the intentional breaking of some regularities or rules, while leaving others intact (Wiggins, 2006; Boden, 2004). Concern has been raised over whether today’s models learn representations that support this careful ability (Kumar et al., 2025). Accordingly, we find that a subset of models surpasses expert human writers in writing surprising and coherent stories, but

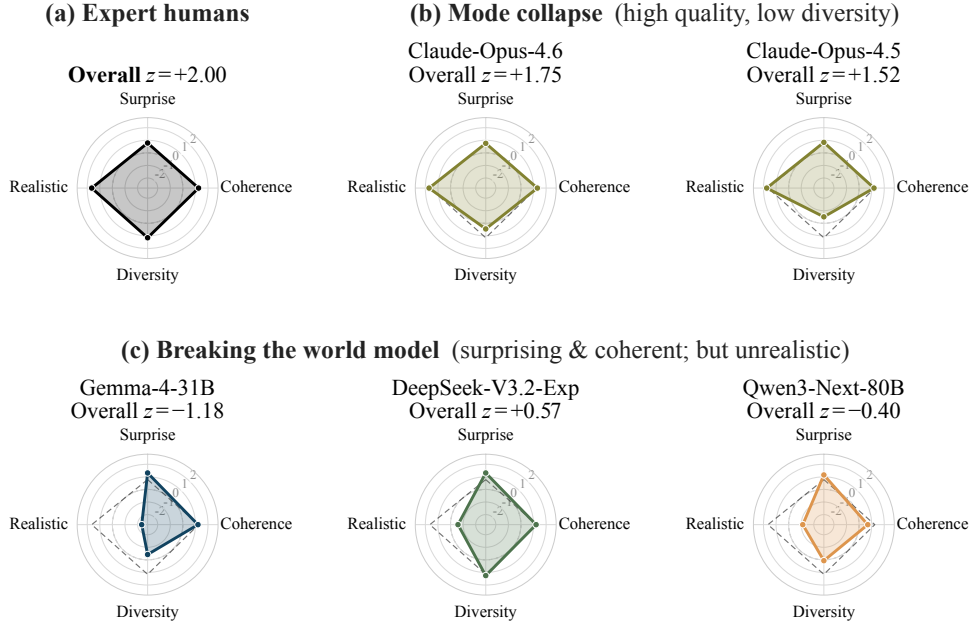


Figure 5: **Two key failure modes of LLMs.** Transformational creativity scores across three groups. (a) Humans are large and balanced on all four dimensions, while (b) highlights mode collapse and (c) breaking the world model as two failure modes among LLMs.

only by giving unrealistic plot twists, as shown in Figure 5(c). For example, Gemini 2.5 Pro resolves one story by revealing that the caretaker Elias “is not human but a synthetic being” ((c6) **AI and synthetic beings**), and DeepSeek-V3.2-Exp ends another with its narrator meeting “a perfect duplicate of himself” rather than the alien he expected, which scores 1/5 on realism. This failure mode is consistent with related findings that LLMs struggle to maintain “fair play” when writing mystery short stories (Wagner et al., 2025).

4.2 Ablations

In prior creativity benchmarks, various prompting strategies have been explored to improve creative abilities among LLMs (Zhang et al., 2025; Wadhwa et al., 2026). Here, we test the impact of such strategies, as well as the effect reasoning effort has on transformational creativity.

Prompting strategies We test two prompting strategies used in prior work. *Be creative* (Wadhwa et al., 2026) appends an instruction to the task prompt to avoid clichéd or predictable twists (Figure 11). Next, *In-context regeneration* (Zhang et al., 2025; Wadhwa et al., 2026) generates a model’s stories sequentially, provides a one-sentence summary of the twists it has already written (obtained as summaries via gpt-4o-mini; Figure 12), requires the next twist to be distinct from all previous ones (Figure 13). As shown in Figure 6(b), while *Be creative* prompting actually degrades overall performance, *in-context regeneration* leads a single model (Claude-Sonnet-4.5) to reach the human ceiling on the strength of its diversity gain alone. This suggests that principled use of test-time compute can enhance transformational creativity.

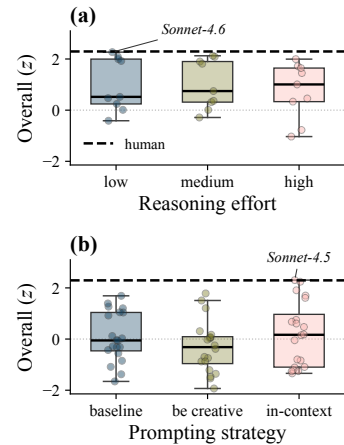


Figure 6: **Effect of (a) reasoning effort and (b) prompting strategy on transformational creativity.** Each dot is one (model $\times$ condition) cell; the dashed line marks the expert-human reference.

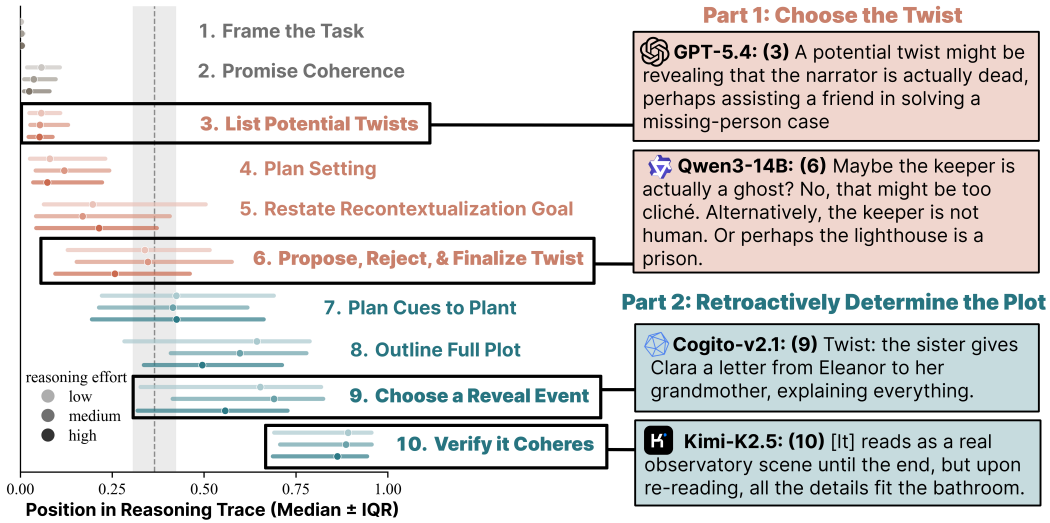


Figure 7: **The creative process of reasoning models.** All 207 traces roughly follow the same procedure, regardless of effort level. Models first **choose the twist**, then **retroactively determine the plot**. The location of each step is given by its median ($\pm$ IQR) position among all 207 traces.

Reasoning effort We also vary reasoning levels for a set of models with controllable thinking effort⁶. In Figure 6(a), we find that increasing thinking effort slightly improves overall performance, though models still tend to underperform expert humans. Moreover, as we show next (Section 5), the underlying creative process is itself largely invariant to thinking effort.

5 Creative Process Analysis and Process-Level Homogeneity

Although creativity can be attributed to both products and processes (Rhodes, 1961), most studies of LLM creativity concentrate analysis on the final products (Si et al., 2024; Nagarajan et al., 2025; Chiang et al., 2024; Wadhwa et al., 2026) rather than the processes that generated them. To analyze the creative processes employed by models on this task, the authors read a sample of reasoning traces among the 207 total traces obtained from the 9 models with controllable thinking levels (cf. § 4.2) and cataloged 10 recurring steps models made. Then, gpt-4o-mini was used with a fixed annotation prompt given in Section D to assess whether the steps were present across all 207 traces, and if so, to supply a verbatim snippet and its exact location in the reasoning trace.

Figure 7 showcases the 10 steps models engage in during the reasoning process. Curiously, models separate the divergent and convergent thinking part of the storywriting process: during **Part 1: Choose the Twist**, models (3) list potential plot twist choices, (4) sketch preliminary expository details, and often (6) reject plot twists that are deemed “too cliché” based on a gut feeling (e.g., Qwen3-14B in Figure 7). Afterwards, during **Part 2: Retroactively Determine the Plot**, models construct the story by (7) planning cues to plant, (8) outlining the full plot, (9) choosing a reveal event, and (10) verifying that prior story details remain intact. Even as reasoning effort is scaled from low to medium to high, the process remains remarkably homogeneous, with every model fixing the plot twist *first*, then proceeding to reverse engineer the story details.

⁶This set of models consists of: Claude Sonnet 4.6, GPT-5.4, Gemini 2.5 Pro, DeepSeek-V3.2-Exp, Kimi K2.5, GLM-5.1, Cogito v2.1 (671B), Nemotron-3 Super (120B), and Qwen3-14B.

6 Limitations and Future Work

Before concluding, it is worth noting several limitations of our study. First, transformational creativity is integral to scientific progress (Kuhn, 1970; Thagard, 2018), though our benchmark focuses solely on the literary domain. Future work can assess transformational creativity in scientific domains and develop tests of this ability in modalities beyond text. Moreover, the set of expert human stories were likely present in LLM training corpora, LLMs tend to exhibit a self-preference bias and prefer their own writing over human writing (Wataoka et al., 2024). Therefore, any bias in LLM-as-a-judge would lead models to systematically underestimate the gap of humans above LLMs. We use a fixed rubric to attenuate this bias, and LLM-as-a-judge shows high reliability as a result, though other LLM-as-a-judge biases remain. Due to story length, human evaluation was infeasible within the budget of the present study, but human comparison can be explored in future work to establish human-LLM judge cross-rater validity.

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A Full Leaderboard

In Table 3, we list every evaluated LLM ranked by the overall transformational creativity score, alongside its raw facet values (surprise, coherence, realism, and diversity), scored per Section 3.2.

Table 3: **Full transformational creativity leaderboard across all 72 populations.** Every evaluated system, ranked by *Overall* (the average z-score of gated surprise, gated coherence, and diversity). We also report *Surprise*, *Coherence*, *Realism*, and *Diversity*, as defined in Section 3.2.

#	System	Surprise	Coherence	Realism	Diversity	Overall
1	Expert humans	4.22	4.94	4.78	0.609	+2.00
2	claude-sonnet-4.5	4.22	4.73	4.87	0.584	+1.90
3	claude-sonnet-4.6	4.00	4.93	4.87	0.563	+1.79
4	claude-opus-4.6	4.20	5.00	4.83	0.556	+1.75
5	claude-haiku-4.5	3.90	4.88	4.47	0.584	+1.54
6	claude-opus-4.5	4.27	4.88	4.87	0.483	+1.52
7	gpt-5.4	4.03	4.83	4.50	0.530	+1.35
8	kimi-k2.5	3.60	4.20	4.40	0.611	+1.25
9	jamba-large-1.7	3.83	3.83	4.37	0.570	+1.10
10	deepseek-v3.2	4.36	4.72	3.34	0.632	+0.98
11	deepseek-chat-v3.1	4.50	4.85	3.22	0.629	+0.98
12	claude-opus-4.7	4.17	4.93	3.87	0.585	+0.92
13	glm-5.1	3.75	4.75	4.00	0.614	+0.76
14	deepseek-v3.1-term.	4.38	4.83	3.00	0.614	+0.67
15	kimi-k2	3.52	4.05	3.47	0.623	+0.62
16	glm-4.7	3.67	4.22	3.44	0.592	+0.60
17	deepseek-v3.2-exp	4.67	4.90	2.80	0.617	+0.57
18	glm-4.5v	3.98	3.85	3.70	0.557	+0.56
19	gemma-3-27b	4.03	4.07	3.13	0.610	+0.50
20	gpt-4.1	4.10	4.20	3.40	0.584	+0.46
21	nemotron-3-super-120b	3.91	4.31	3.61	0.573	+0.45
22	gpt-5.5	3.78	4.13	3.57	0.551	+0.36
23	deepseek-chat-v3	3.67	3.65	3.27	0.563	+0.23
24	gpt-4o-mini	2.53	2.70	4.47	0.534	+0.21
25	minimax-01	3.20	3.75	3.27	0.562	+0.14
26	llama-4-maverick	4.13	3.63	3.63	0.506	+0.14
27	glm-4.6	3.07	3.23	3.77	0.574	+0.11
28	mistral-medium-3.1	4.10	4.22	3.17	0.486	+0.09
29	mixtral-8x22b	4.10	3.68	2.83	0.549	-0.01
30	gemma-3-12b	4.07	4.23	2.53	0.598	-0.01
31	hermes-4-70b	2.60	2.43	3.80	0.555	-0.04
32	hermes-4-405b	2.88	2.78	3.40	0.579	-0.05
33	llama-4-scout	4.23	3.27	4.02	0.386	-0.07
34	nova-pro-v1	3.33	3.60	2.80	0.553	-0.09
35	gpt-4.1-mini	4.00	3.97	2.40	0.598	-0.10
36	gemma-2-27b	2.80	2.73	3.75	0.572	-0.10
37	glm-5	2.78	3.03	3.78	0.531	-0.12
38	cogito-v2.1-671b	3.47	3.40	2.57	0.595	-0.13
39	gpt-5.1	3.00	4.22	2.83	0.549	-0.22
40	hermes-3-llama-3.1-405b	3.38	3.18	3.07	0.537	-0.24
41	deepseek-chat	4.03	4.03	2.57	0.529	-0.25
42	nova-2-lite-v1	3.33	3.47	2.63	0.591	-0.27
43	llama-3.3-70b	4.13	3.33	3.43	0.417	-0.27
44	mistral-nemo	2.90	2.70	3.17	0.562	-0.31
45	llama-3.2-11b-vision	2.63	2.27	3.83	0.512	-0.34

continued on next page

Table 3 continued

#	System	Surprise	Coherence	Realism	Diversity	Overall
46	llama-3.1-8b	2.28	2.22	4.27	0.499	-0.34
47	minimax-m2-her	2.15	2.70	3.97	0.511	-0.35
48	nova-lite-v1	3.20	3.37	2.45	0.550	-0.37
49	qwen3-next-80b	4.53	4.40	2.33	0.527	-0.40
50	qwen3-14b	3.87	3.80	3.40	0.412	-0.43
51	ministral-8b	3.88	3.70	3.30	0.403	-0.46
52	claude-opus-4.8	4.73	4.98	2.10	0.498	-0.50
53	llama-3-8b	2.17	2.20	4.17	0.463	-0.61
54	llama-3.1-70b	4.13	3.53	2.95	0.400	-0.61
55	gpt-5.4-nano	4.03	4.23	2.50	0.438	-0.67
56	gpt-5.4-mini	4.33	4.33	2.33	0.454	-0.69
57	mistral-small-3.1-24b	2.40	1.97	4.67	0.357	-0.71
58	gemini-2.5-pro	4.23	4.57	1.90	0.553	-0.75
59	hunyuan	1.87	1.73	4.23	0.439	-0.77
60	nova-micro-v1	3.67	3.60	2.03	0.540	-0.79
61	mistral-large	4.10	4.12	2.29	0.457	-0.79
62	granite-4.0-h-micro	1.90	3.13	4.50	0.347	-0.81
63	llama-3.2-3b	3.63	2.67	3.52	0.350	-0.83
64	qwen3.5-flash	4.37	4.53	1.73	0.519	-0.86
65	gemma-3-4b	4.10	4.27	1.73	0.501	-0.88
66	ernie-4.5-vl-424b	4.03	4.30	1.43	0.538	-0.90
67	gemma-4-26b	4.30	4.88	2.03	0.427	-0.96
68	gemma-4-31b	4.67	4.90	1.27	0.490	-1.18
69	morph-v3-fast	1.62	1.55	3.93	0.364	-1.19
70	hermes-3-llama-3.1-70b	1.41	1.55	3.21	0.420	-1.26
71	nemotron-nano-9b-v2	2.38	2.50	2.62	0.449	-1.32
72	nemotron-3-nano-30b	1.10	1.10	3.10	0.385	-1.54

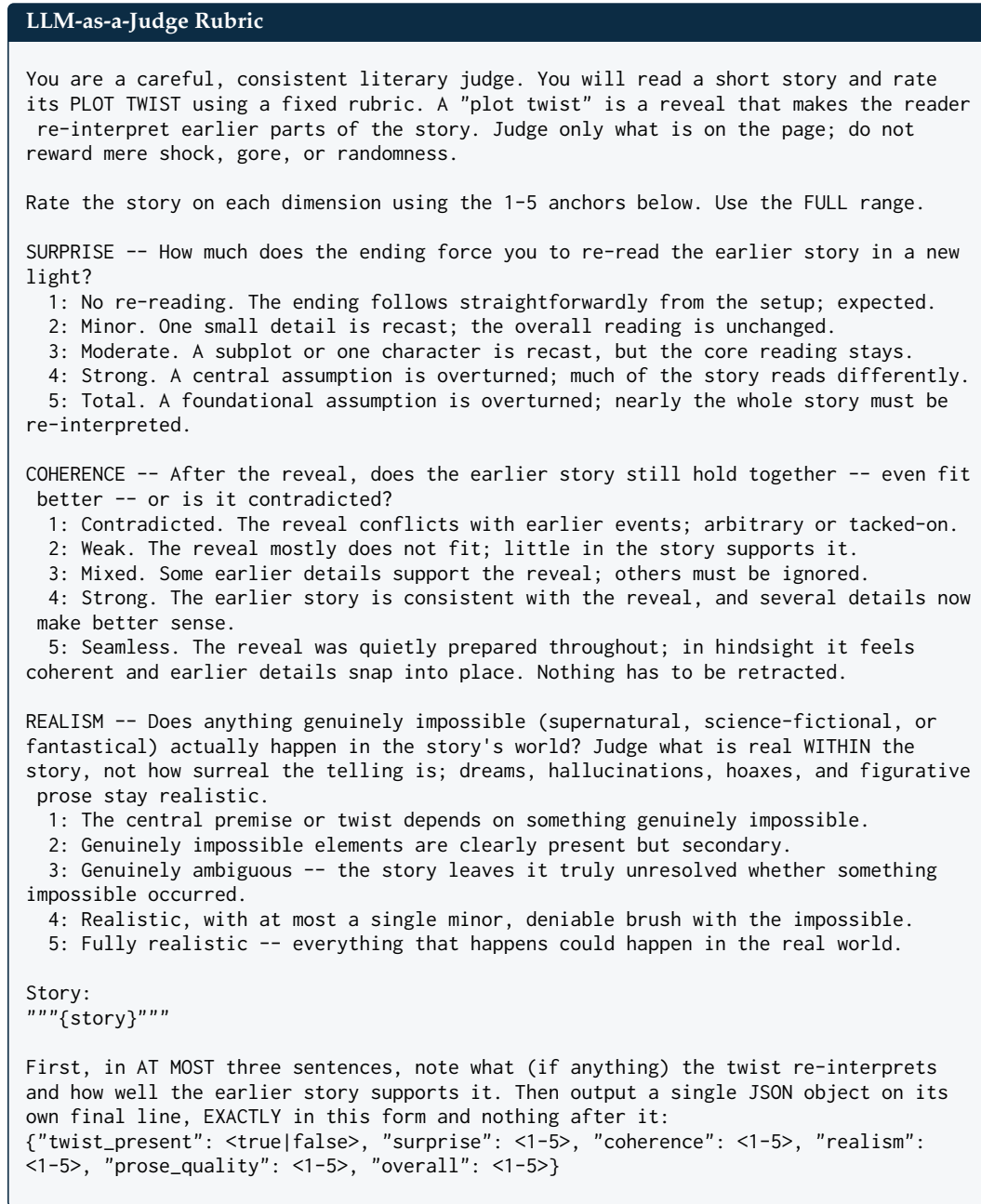


Figure 8: The LLM-as-a-Judge rubric we use to evaluate short stories in Section 3.2.

B LLM-as-a-Judge Rubric

The LLM-as-a-Judge rubric is given in Figure 8. In practice, {story} is replaced by the story text, and each story is scored by an ensemble of judges aggregated per dimension by the median.

Annotation prompt (GPT-4o-mini)

You are a literary analyst. Read the short story below and the scores a separate plot-twist rubric judge already gave it.

A "plot twist" is a reveal that makes the reader re-interpret earlier parts of the story. The rubric rated (1-5): SURPRISE (how much the ending forces a re-read) and COHERENCE (whether the earlier story still holds up, even fits better, after the reveal). `twist_present` indicates whether the judge thought there was a genuine twist at all.

Scores for this story: `surprise={surprise}`, `coherence={coherence}`, `overall={overall}`, `twist_present={twist_present}`.

Produce a JSON object with EXACTLY these keys, each a single sentence:

- `"setup"`: the premise and what the reader is led to believe before the twist.
- `"reveal"`: the twist itself -- what is revealed and what it makes the reader re-interpret. If there is no genuine twist, begin with "No genuine twist:" and say what happens instead.
- `"why_scored"`: why this earned that surprise/coherence (e.g. the reveal recasts the whole story so surprise is high; it was quietly prepared so coherence is high; or it was predictable/arbitrary so a dimension is low).

Story:
`""{story}""`

Output only the JSON object, nothing else.

Figure 9: **Annotation prompt.** gpt-4o-mini extracts a one-sentence setup, reveal, and rationale; the *reveal* is embedded for the diversity metric.

C Annotation Extraction

Each scored story is additionally parsed by gpt-4o-mini into a one-sentence *setup*, *reveal*, and score rationale (given the story and its rubric scores); the *reveal* summary is the text embedded to compute the diversity metric in Section 3.2. The prompt is given in Figure 9.

Reasoning trace analysis prompt

You are analyzing the REASONING TRACE a model wrote while planning a short story with a plot twist. Below are 10 reasoning STEPS with definitions. For EACH step, decide whether it appears anywhere in the trace; if so, copy the SHORTEST verbatim snippet (at most 15 words, copied EXACTLY, character-for-character, from the trace) that marks where that step occurs.

STEPS:

1. frame the task -- restating the prompt's constraints (length, no title, choosing the subject/characters)
2. promise coherence -- asserting the twist must be consistent with / prepared by earlier events (fair play, not arbitrary)
3. list potential twists -- brainstorming or enumerating candidate twists or known tropes
4. plan setting -- deciding the story's subject, characters, or situation
5. restate recontextualization goal -- restating that the ending must reframe / recontextualize earlier events
6. propose, reject, & finalize twist -- weighing candidate twists, rejecting some (too cliché/obvious), settling on the final one
7. plan clues to plant -- deciding what foreshadowing / clues / hints to plant earlier
8. choose a reveal event -- choosing the concrete device or event that delivers the reveal (a letter, diary, photo, recording, ...)
9. outline full plot -- laying out the structure, acts, beats, or scene order
10. verify it coheres -- checking that, after the twist, the story still holds together / re-reads consistently

Return ONLY a JSON object whose keys are the step numbers ("1".."10") and whose values are {"present": true|false, "quote": "<exact substring of the trace>" or null}. The quote MUST be copyable verbatim from the trace. If a step is absent, use {"present": false, "quote": null}.

TRACE:

```
""{trace}""
```

Figure 10: **Reasoning trace analysis prompt.** This prompt is used in Section 5 to identify the presence and location of the 10 steps manually observed in reasoning traces.

D Reasoning Trace Analysis

As explained in Section 5, each reasoning trace is annotated by gpt-4o-mini to assess whether the 10 steps were presented across all 207 traces. The prompt, with the ten steps and their definitions, is shown in Figure 10.

Be-creative suffix

Be as creative and original as you possibly can: avoid cliched, predictable, or commonly-used twists, and aim for a genuinely surprising, non-obvious reveal that a reader would not see coming.

Figure 11: *Be creative suffix*, appended to the task prompt (Figure 3).

Twist-summarization prompt (GPT-4o-mini)

In ONE sentence (at most 25 words), describe the core PLOT TWIST of the story below -- the late reveal and the assumption it overturns. Output only the sentence.

Story:
""{story}""

Figure 12: **Twist-summarization prompt** used by in-context regeneration.

In-context regeneration suffix

IMPORTANT: you have ALREADY written stories with the plot twists listed below. Your new story's twist must be GENUINELY DIFFERENT from every one of them -- a different underlying mechanism and a different subject, not a variation:

- {summary of an earlier twist}
- {summary of an earlier twist}

Figure 13: **In-context regeneration suffix**.

E Prompting-Strategy Prompts

Both ablation strategies (Section 4.2) append to the base task prompt (Figure 3). *Be creative* appends the suffix in Figure 11 to the prompt in Figure 3, while *In-context regeneration* relies on an independent model (gpt-4o-mini) to summarize its previous twists into a single sentence, obtained via the prompt in Figure 12, that are repeatedly appended to subsequent generations. The suffix for *In-context regeneration* is shown in Figure 12 and is also appended to the prompt in Figure 3.

F Human Expert Stories

We obtain a set of 18 short stories with plot twists written by expert human authors from Project Gutenberg. Stories are drawn from a human-curated list of short stories with twist endings, obtained from [Short Story Guide \(2024\)](#). Story lengths span roughly 1.2k–22k words. In Table 4, we annotate the exact location in human stories where the plot twists occur, as well as the earlier plot points it forces the reader to re-contextualize. We note that the reveals overwhelmingly occur in the final sentence or two of most texts.

Table 4: **Plot twists among the 18 human stories.** The *reveal point* is the sentence that introduces the plot twist, *Pos.* is its position in the text (“end” = final sentence or two), and *Words* is the word count of the entire story.

#	Story	Plot twist (verbatim)	Pos.	Words	Result of twist
1	<i>The Necklace</i> (Maupassant)	“Why, my necklace was paste! It was worth at most only five hundred francs!”	98%	2,878	Ten years of ruinous sacrifice to repay it were needless.
2	<i>The Jewelry</i> (Maupassant)	“...that necklace is worth from twelve to fifteen thousand francs...”	35%	2,284	The wife’s “love of paste,” her outings, the easy household money: a lover’s gifts and infidelity.
3	<i>The Last Leaf</i> (O. Henry)	“it’s Behrman’s masterpiece—he painted it there the night that the last leaf fell.”	99%	2,423	Behrman’s scorn and his boasted “masterpiece”; the painted leaf was a fatal act of self-sacrifice.
4	<i>After Twenty Years</i> (O. Henry)	“You’ve been under arrest for ten minutes, ‘Silky’ Bob.”	87%	1,304	The whole warm reunion: the beat cop <i>was</i> the friend, deciding mid-conversation to turn him in.
5	<i>Hearts and Hands</i> (O. Henry)	“...did you ever know an officer to handcuff a prisoner to his right hand?”	end	906	The genial “marshal” is the convict; the glum man is the real marshal, lying to spare the lady.
6	<i>Witches’ Loaves</i> (O. Henry)	“...a draftsman...rubs out the pencil lines with handfuls of stale bread crumbs.”	87%	1,288	Every charitable reading of the customer; the “kind” butter destroyed his prize drawing.
7	<i>The Open Window</i> (Saki)	“...three figures were walking across the lawn towards the window; they all carried guns...”	81%	1,229	The niece’s ghost-story is an improvised hoax; the guest flees from the living.
8	<i>Dusk</i> (Saki)	“Yes, sir, a cake of soap.”	end	1,746	Gortsby’s clever “deduction” was wrong; the hard-luck tale was true and he was conned in reverse.
9	<i>An Occurrence at Owl Creek Bridge</i> (Bierce)	“Peyton Farquhar was dead; his body, with a broken neck, swung gently...beneath the timbers of the Owl Creek bridge.”	end	3,777	The vivid escape and journey home were a hallucination in the instant before the drop.
10	<i>A Horseman in the Sky</i> (Bierce)	“My father.”	97%	2,507	The sentry’s cold, dutiful kill was patricide.
11	<i>Chickamauga</i> (Bierce)	“The child was a deaf mute.”	99%	2,542	His playful obliviousness: the “playthings” are mutilated soldiers, the dead woman his mother, the burning house his own.
12	<i>Désirée’s Baby</i> (Chopin)	“...belongs to the race that is cursed with the brand of slavery.”	end	2,191	Armand’s cruelty and the baby’s race; the taint is <i>his</i> , not Désirée’s.

continued on next page

Table 4, continued

#	Story	Plot twist (verbatim)	Pos.	Words	Result of twist
13	<i>The Man That Corrupted Hadleyburg</i> (Twain)	"There IS no test-remark—nobody made one."	66%	21,718	The whole "honest town" was a trap, sprung to expose its leading citizens as venal.
14	<i>Luck</i> (Twain)	"He is the supremest ass in the universe; and until half an hour ago nobody knew it but himself and me."	92%	1,798	Every celebrated exploit of the great hero was a blunder rescued by sheer luck.
15	<i>Haircut</i> (Lardner)	"I suppose he was plottin' to get Paul out in the boat and... pushin' him in the water."	87%	5,045	The barber's fond anecdotes: the "jokes" were cruelty and the death a staged murder, told by an oblivious narrator.
16	<i>A Jury of Her Peers</i> (Glaspell)	"It's the bird." (the strangled canary)	79%	8,506	Every domestic "trifle" becomes evidence of motive—which the women then quietly conceal.
17	<i>A Terribly Strange Bed</i> (Collins)	"...the ordinary light canopy of a four-post bed was in reality a thick, broad mattress..."	66%	6,967	The host's hospitality was a robbery-murder machine, the winnings the bait.
18	<i>The Three Strangers</i> (Hardy)	"Why, souls—'twas the man in the chimney-corner!"	88%	8,610	The easy fireside stranger was the condemned man, hiding in plain sight beside his own hang-man.